

ENTERPRISE SOC MONITORING LAB

DoS/DDoS Detection & Incident Response

Security Operations Center — Home Lab Project

Field

Analyst
Role
Lab Environment
Domain
Date
MITRE ATT&CK Version
Tools

Details

Tem Elvis
SOC Analyst
VirtualBox — Home SOC Lab
cis.net
June 21, 2026
v14
pfSense | Snort IDS | Splunk SIEM | hping3 | Nmap | Wireshark

1. Executive Summary

This report documents the full lifecycle of a DoS/DDoS attack simulation conducted within a purpose-built home Security Operations Center (SOC) lab environment. The exercise covered reconnaissance, attack simulation, real-time detection, SIEM visualization, automated alerting, and structured incident response — all mapped to the MITRE ATT&CK framework.

Three distinct attack types were simulated from an isolated Kali Linux attacker machine (192.168.2.5) targeting a Windows Server 2025 domain controller (192.168.1.1) running Microsoft IIS on port 80. Custom Snort IDS rules detected all attacks in real time, with events forwarded to Splunk Enterprise for correlation, dashboard visualization, and automated alerting.

Key Findings:

- 1,528,725 total security events detected and indexed in Splunk during the simulation period
- 4 active DoS/DDoS attack sessions identified and classified at HIGH severity
- 138,186 SYN flood packets transmitted causing 100% packet loss on the target
- 144,995 HTTP flood detection events generated in Splunk
- 610,013 UDP flood events captured within a 15-minute window
- Splunk automated alerts (DOS Alerts) triggered 18+ times between 17:05 and 21:40 WAT
- Snort IDS detected reconnaissance activity (15 ICMP PING NMAP alerts) before any attack launched
- Wireshark packet capture confirmed SYN flood behavior and TCP error patterns at network level
- pfSense firewall logged all inter-zone traffic, providing perimeter-level visibility

This project demonstrates a practical, layered defense architecture — pfSense perimeter filtering, Snort signature-based detection, and Splunk centralized SIEM — directly applicable to enterprise SOC analyst roles.

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2. Network Topology & Lab Architecture

2.1 Architecture Overview

The lab environment was built using Oracle VirtualBox with five virtual machines communicating across four isolated network segments. pfSense acts as the central firewall, router, and DHCP server enforcing traffic policies between all zones.

2.2 Network Diagram

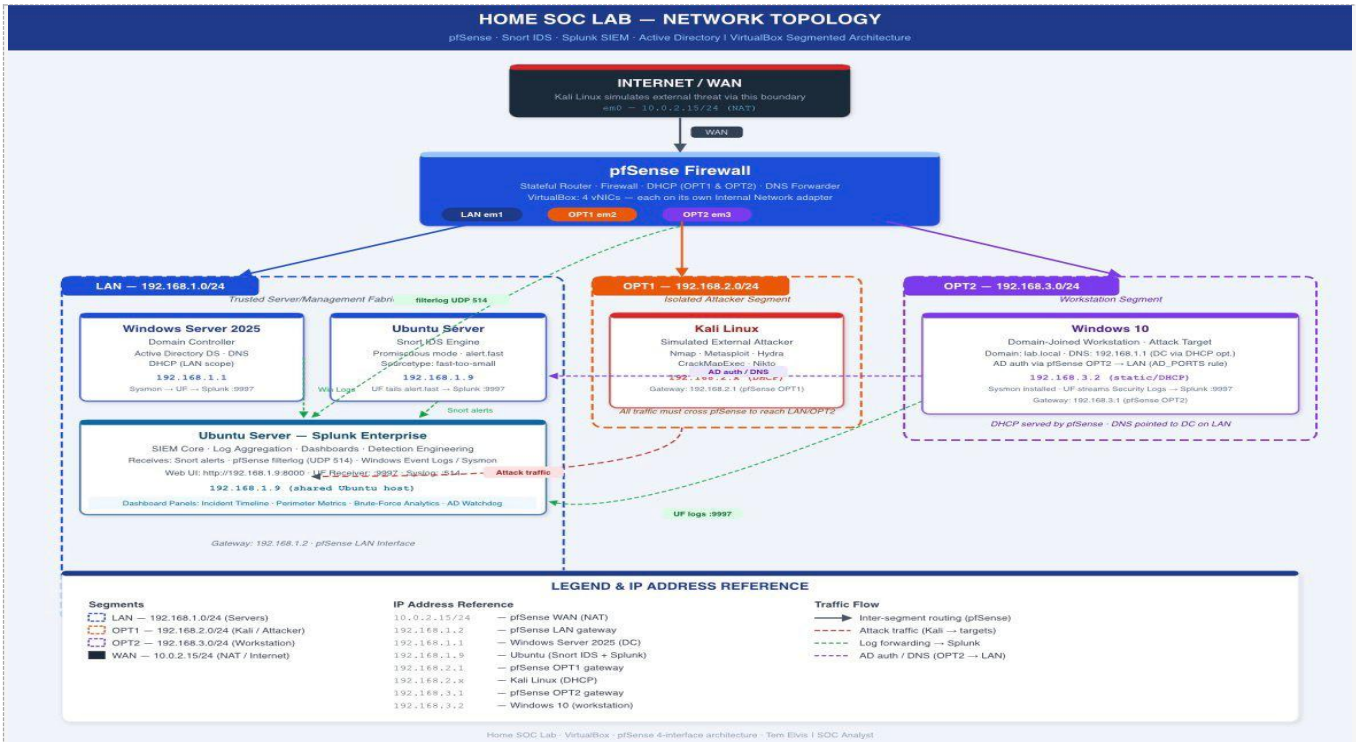
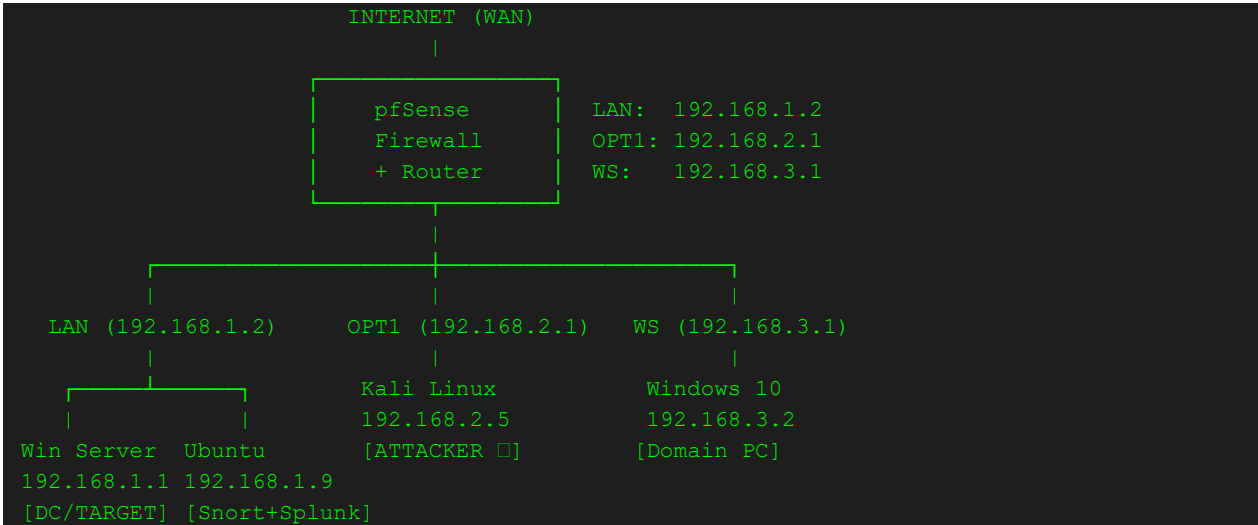


Figure 2.1: Home SOC Lab Network Topology — 4-zone segmented architecture

2.3 Device Inventory

Device	OS	IP Address	Role
pfSense	FreeBSD	192.168.1.2 / 192.168.2.1 / 192.168.3.1	Firewall / Router / DHCP
Windows Server 2025	Windows Server	192.168.1.1	Domain Controller / IIS Target
Ubuntu Server	Ubuntu Linux	192.168.1.9	Snort IDS + Splunk SIEM
Windows 10	Windows 10	192.168.3.2	Domain Workstation (DHCP)
Kali Linux	Debian Linux	192.168.2.5	Attacker Machine ☐

2.4 Network Zones

Interface	Subnet	Zone	Trust Level	Devices
WAN	Dynamic	Internet	Untrusted	External Internet
LAN	192.168.1.0/24	Server Zone	Trusted	Windows Server, Ubuntu
OPT1	192.168.2.0/24	Attacker Zone	Untrusted	Kali Linux
Workstation	192.168.3.0/24	User Zone	Semi-Trusted	Windows 10

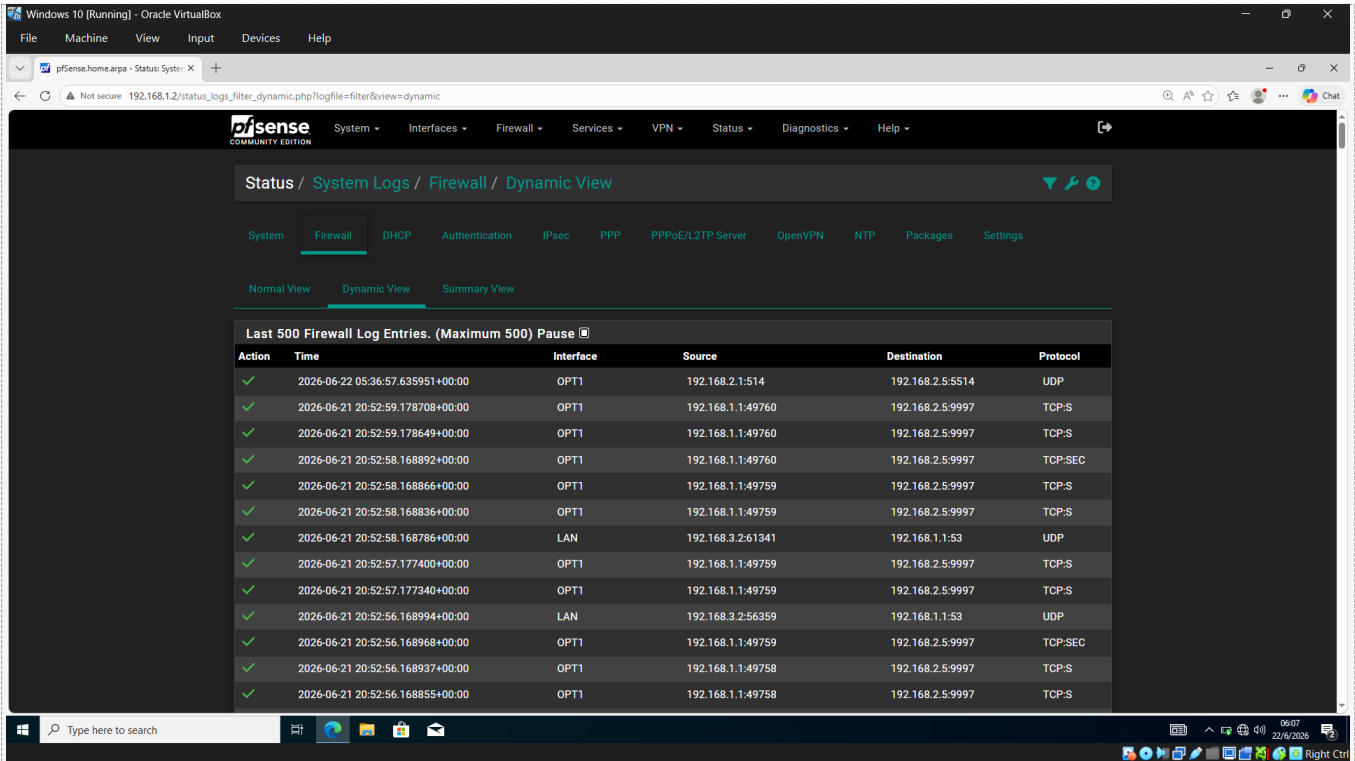


Figure 2.2: pfSense Firewall Dashboard showing interface status and traffic

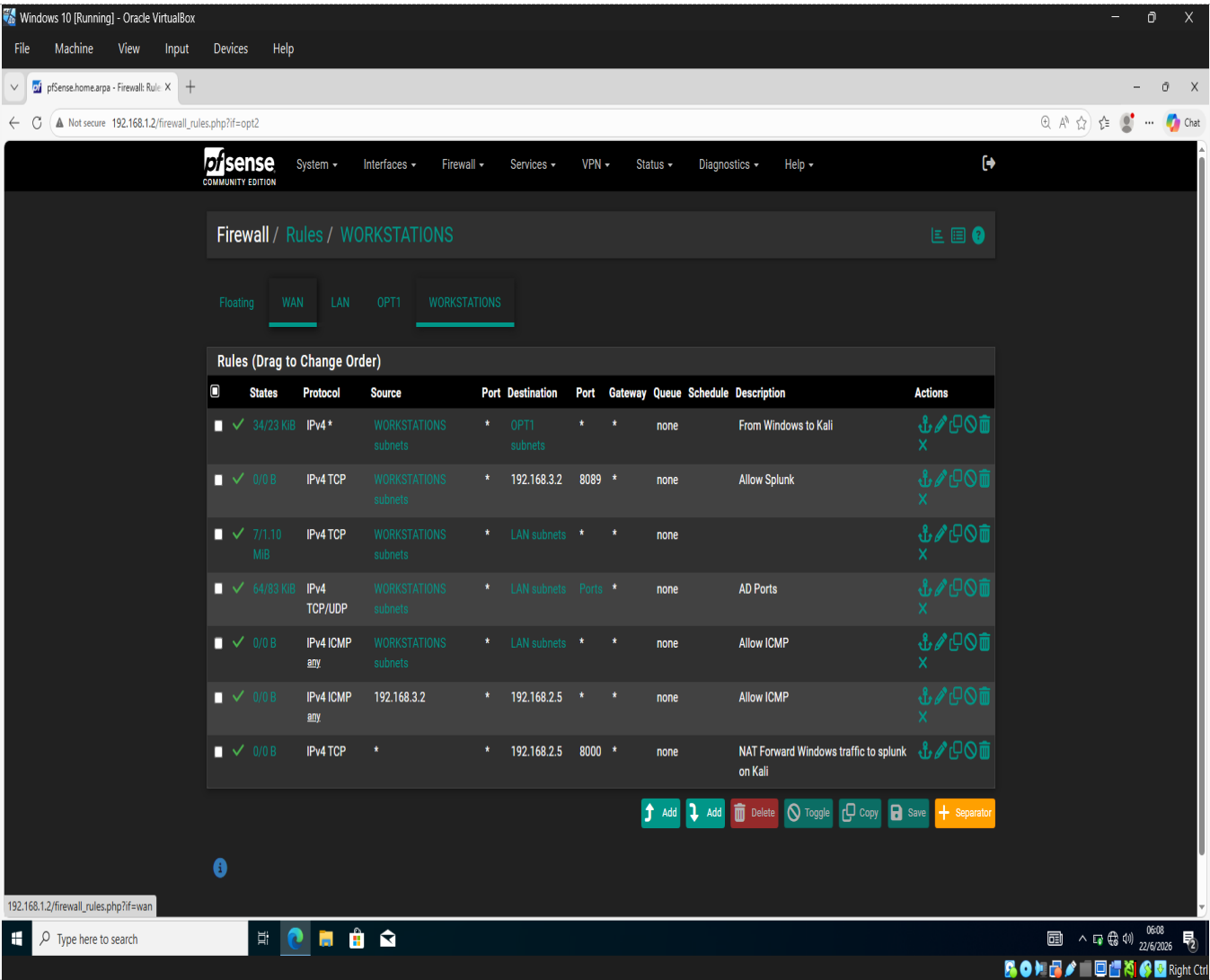


Figure 2.3: pfSense Firewall Rules enforcing zone-based traffic policies

3. Detection Stack & Log Pipeline

Layer	Tool	Version	Role
Perimeter Defense	pfSense	CE Latest	Firewall rules, traffic logging, DHCP
Network IDS	Snort	3.x	Signature-based intrusion detection
Log Forwarding	Splunk Universal Forwarder	9.x	Ship logs from Ubuntu to Splunk
SIEM	Splunk Enterprise	10.2.2	Log aggregation, dashboards, alerting
Packet Analysis	Wireshark	Latest	Deep packet inspection
Attack Tool	hping3	Latest	DoS/DDoS simulation
Reconnaissance	Nmap	7.9x	Network scanning and service detection

3.1 Log Flow Architecture

```
Snort IDS (Ubuntu 192.168.1.9)
└─ /var/log/snort/snort.alert.fast [plain text alert_fast format]
   └─ Splunk UF inputs.conf monitors this file
      └─ index=security sourcetype=fast_alert
         └─ SOC Dashboard panels + DOS Alerts
```

```
pfSense Firewall
└─ Syslog UDP:5514 → Ubuntu 192.168.1.9
   └─ index=pfsense sourcetype=syslog
      └─ Perimeter Firewall Dashboard Panel
```

```
Windows Server / Windows 10
└─ Splunk UF → WinEventLog:Security
   └─ index=* sourcetype=WinEventLog:Security
      └─ Brute Force + AD Monitoring Panels
```

3.2 Snort Configuration

Snort was configured to write alerts in alert_fast plain text format to /var/log/snort/snort.alert.fast. This required adding the following output directive to /etc/snort/snort.conf:

```
output alert_fast: alert
```

The Splunk Universal Forwarder inputs.conf was configured to monitor this file:

```
[monitor:///var/log/snort/snort.alert.fast]
disabled = false
index = security
sourcetype = fast_alert
crcSalt = <SOURCE>
```


4. Snort Detection Rules

Eight custom Snort rules were written and deployed to `/etc/snort/rules/local.rules` covering all simulated attack types. Each rule is mapped to a MITRE ATT&CK technique ID and uses the `detection_filter` keyword to prevent false positives from low-volume traffic.

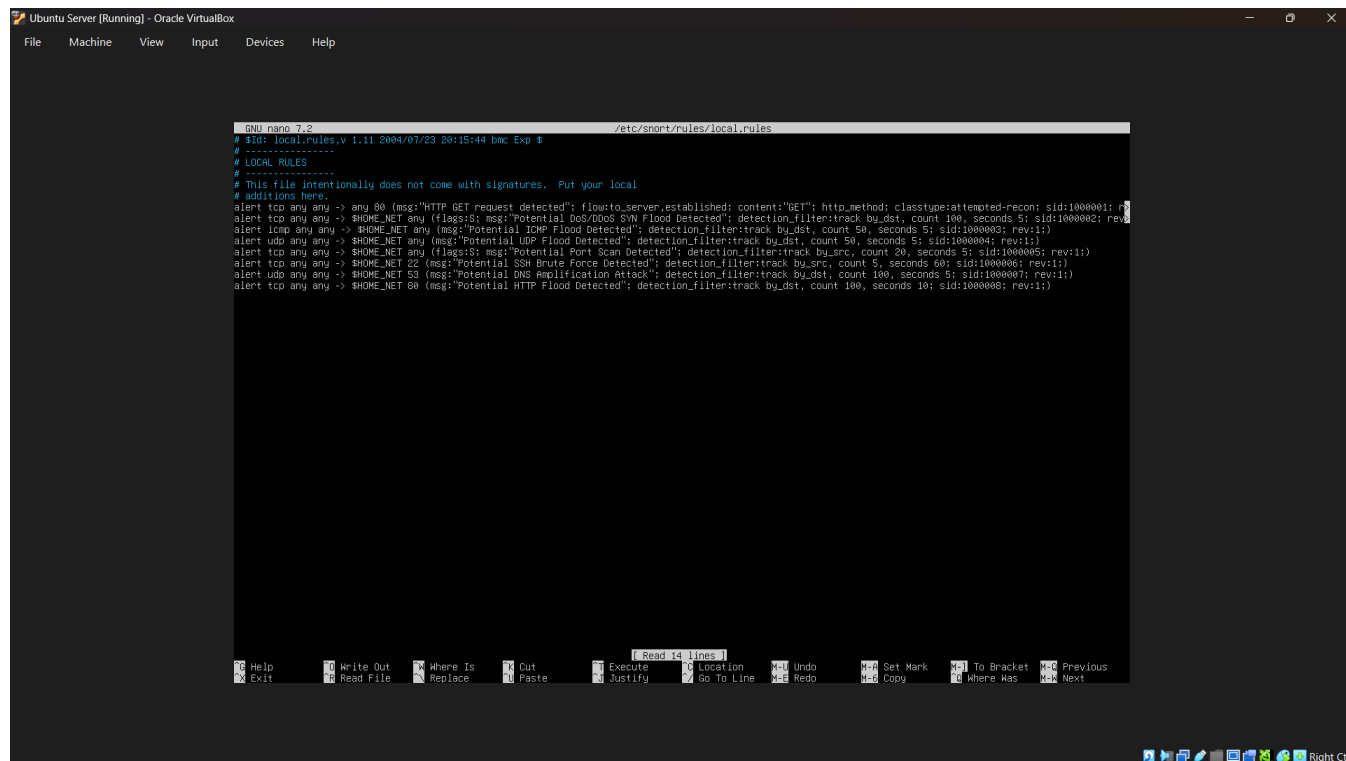


Figure 4.1: Custom Snort detection rules in `/etc/snort/rules/local.rules`

4.1 Rule Definitions

```
# HTTP GET Detection - T1595
alert tcp any any -> any 80 (msg:"HTTP GET Request Detected";
    flow:to_server,established; content:"GET"; http_method;
    classtype:attempted-recon; sid:1000001; rev:1;)
```

```
# SYN Flood Detection - T1499.001
alert tcp any any -> $HOME_NET any (flags:S;
    msg:"Potential DoS/DDoS SYN Flood Detected";
    detection_filter:track by_dst, count 100, seconds 5;
    sid:1000002; rev:1;)
```

```
# ICMP Flood Detection - T1498
alert icmp any any -> $HOME_NET any
```

```
(msg:"Potential ICMP Flood Detected";  
detection_filter:track by dst, count 50, seconds 5; sid:1000003; rev:1;)
```

```
# UDP Flood Detection - T1498  
alert udp any any -> $HOME_NET any  
(msg:"Potential UDP Flood Detected";  
detection_filter:track by dst, count 50, seconds 5; sid:1000004; rev:1;)
```

```
# Port Scan Detection - T1595.001  
alert tcp any any -> $HOME_NET any (flags:S;  
msg:"Potential Port Scan Detected";  
detection_filter:track by src, count 20, seconds 5; sid:1000005; rev:1;)
```

```
# SSH Brute Force - T1110  
alert tcp any any -> $HOME_NET 22  
(msg:"Potential SSH Brute Force Detected";  
detection_filter:track by src, count 5, seconds 60; sid:1000006; rev:1;)
```

```
# DNS Amplification - T1498.002  
alert udp any any -> $HOME_NET 53  
(msg:"Potential DNS Amplification Attack";  
detection_filter:track by dst, count 100, seconds 5; sid:1000007; rev:1;)
```

```
# HTTP Flood Detection - T1499.002  
alert tcp any any -> $HOME_NET 80  
(msg:"Potential HTTP Flood Detected";  
detection_filter:track by dst, count 100, seconds 10; sid:1000008; rev:1;)
```

4.2 Rule Summary

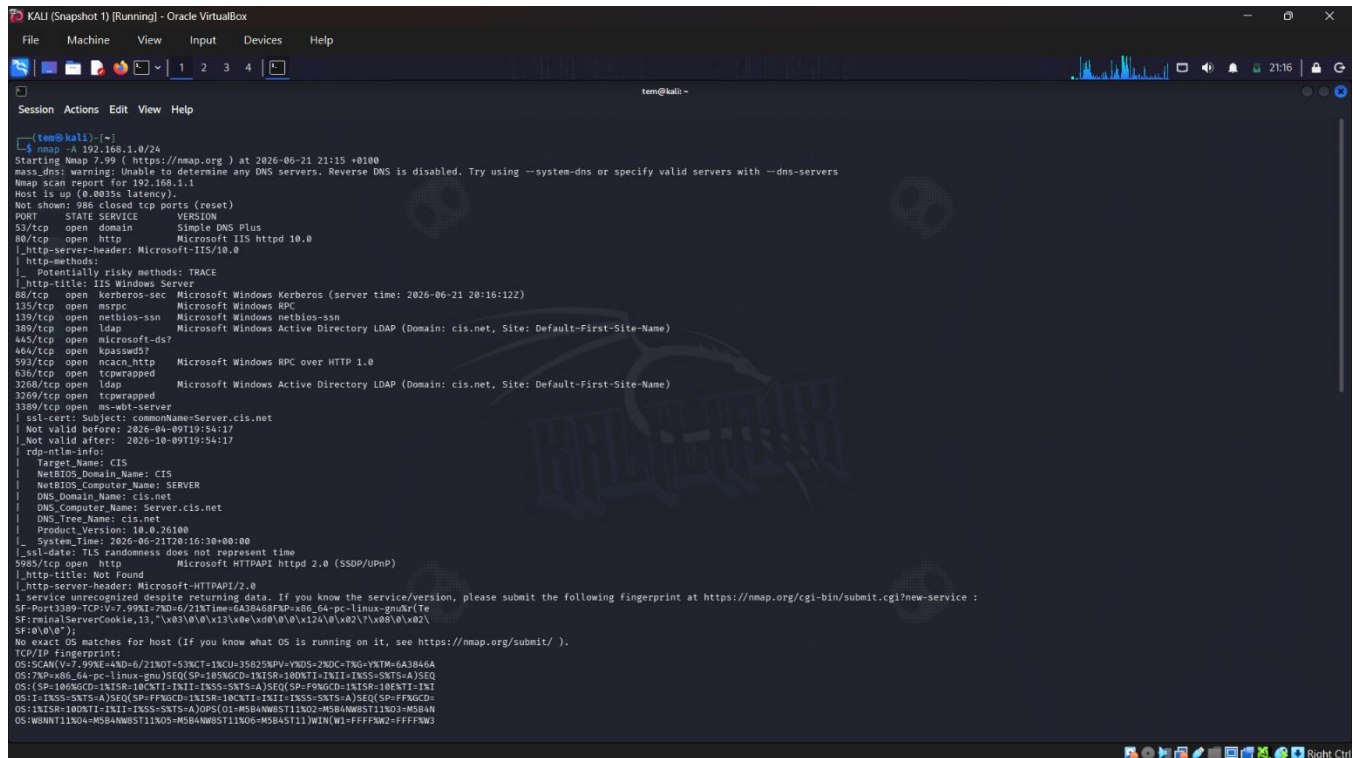
SID	Rule Name	Protocol	MITRE ID	Threshold
1000001	HTTP GET Request	TCP	T1595	N/A
1000002	SYN Flood	TCP	T1499.001	100 pkts / 5s per dst
1000003	ICMP Flood	ICMP	T1498	50 pkts / 5s per dst
1000004	UDP Flood	UDP	T1498	50 pkts / 5s per dst
1000005	Port Scan	TCP	T1595.001	20 pkts / 5s per src
1000006	SSH Brute Force	TCP	T1110	5 pkts / 60s per src
1000007	DNS Amplification	UDP	T1498.002	100 pkts / 5s per dst
1000008	HTTP Flood	TCP	T1499.002	100 pkts / 10s per dst

5. Phase 1 — Reconnaissance (MITRE T1595)

Before launching any attack, active reconnaissance was performed from Kali Linux (192.168.2.5) targeting the LAN segment (192.168.1.0/24). This mirrors real-world attacker behavior where targets are identified and profiled before an attack is launched.

5.1 Network Scan

```
nmap -A 192.168.1.0/24
```



```
(tem@kali)~$ nmap -A 192.168.1.0/24
Starting Nmap 7.99 ( https://nmap.org ) at 2026-06-21 21:15 +0100
mass_dns: warning: Unable to determine any DNS servers. Reverse DNS is disabled. Try using --system-dns or specify valid servers with --dns-servers
Nmap scan report for 192.168.1.1
Host is up (0.003s latency).
Not shown: 986 closed tcp ports (reset)
PORT      STATE SERVICE
53/tcp    open  domain
80/tcp    open  http
|_http-server-header: Microsoft-IIS/10.0
|_http-methods:
|_http-title: IIS Windows Server
88/tcp    open  kerberos-sec
135/tcp   open  msrpc
139/tcp   open  netbios-ssn
389/tcp   open  ldap
445/tcp   open  microsoft-ds
464/tcp   open  kpasswst
593/tcp   open  ncacn_http
636/tcp   open  tcpwrapped
3268/tcp  open  ldap
3269/tcp  open  tcpwrapped
3389/tcp  open  ms-wbt-server
|_ssl-cert: Subject: commonName=Server.cis.net
|_Not valid before: 2026-04-09T19:54:17
|_Not valid after: 2026-10-09T19:54:17
|_rdp-ntlm-info:
|_Target_Name: CIS
|_NetBIOS_Domain_Name: CIS
|_NetBIOS_Computer_Name: SERVER
|_DNS_Domain_Name: cis.net
|_DNS_Computer_Name: Server.cis.net
|_DNS_Tree_Name: cis.net
|_Product_Version: 10.0.26100
|_System_Time: 2026-06-21T20:16:30+00:00
|_ssl-date: TLS randomness does not represent time
5985/tcp  open  http
|_http-title: Not found
|_http-server-header: Microsoft-HTTPAPI/2.0
I service unrecognized despite returning data. If you know the service/version, please submit the following fingerprint at https://nmap.org/cgi-bin/submit.cgi?new-service :
SF-Port3389-TCP-V:7.99RI:72D-6/21NTIME:6A3B46FAP:x86_64-pc-linux-gnuX(Te
SF:rminalServerCookie,13,"\\x03\\x00\\x13\\x0e\\xd0\\x00\\x124\\x02\\x08\\x02\\
SF:x00\\x00");
No exact OS matches for host (If you know what OS is running on it, see https://nmap.org/submit/ ).
TCP/IP fingerprint:
OS:SCAN(V=7.99NE=4ND=6/21NOT=53NCT=1NCU=358258PV=YKDS=2MDC=TWG=YKTM=6A3B46A
OS:7RP=x86_64-pc-linux-gnu)SEQ(SP=185NGCD=1X1SR=10DXTI=1X1I=INSS=5NTS=A)SEQ
OS:(CP=100NGCD=1X1SR=10DXTI=1X1I=INSS=5NTS=A)SEQ(SP=FFNGCD=1X1SR=10DXTI=1X1
OS:I=INSS=5NTS=A)SEQ(SP=FFNGCD=1X1SR=10DXTI=1X1I=INSS=5NTS=A)SEQ(SP=FFNGCD=
OS:I=INSS=10DXTI=1X1I=INSS=5NTS=A)OPS(OI=MSB4NWBST11N02=MSB4NWBST11N03=MSB4N
OS:W0NNF11N04=MSB4NWBST11N05=MSB4NWBST11N06=MSB4S111NIN(WI=FFFFN2=FFFFN3
```

Figure 5.1: Nmap aggressive scan revealing Windows Server 2025 services on 192.168.1.1

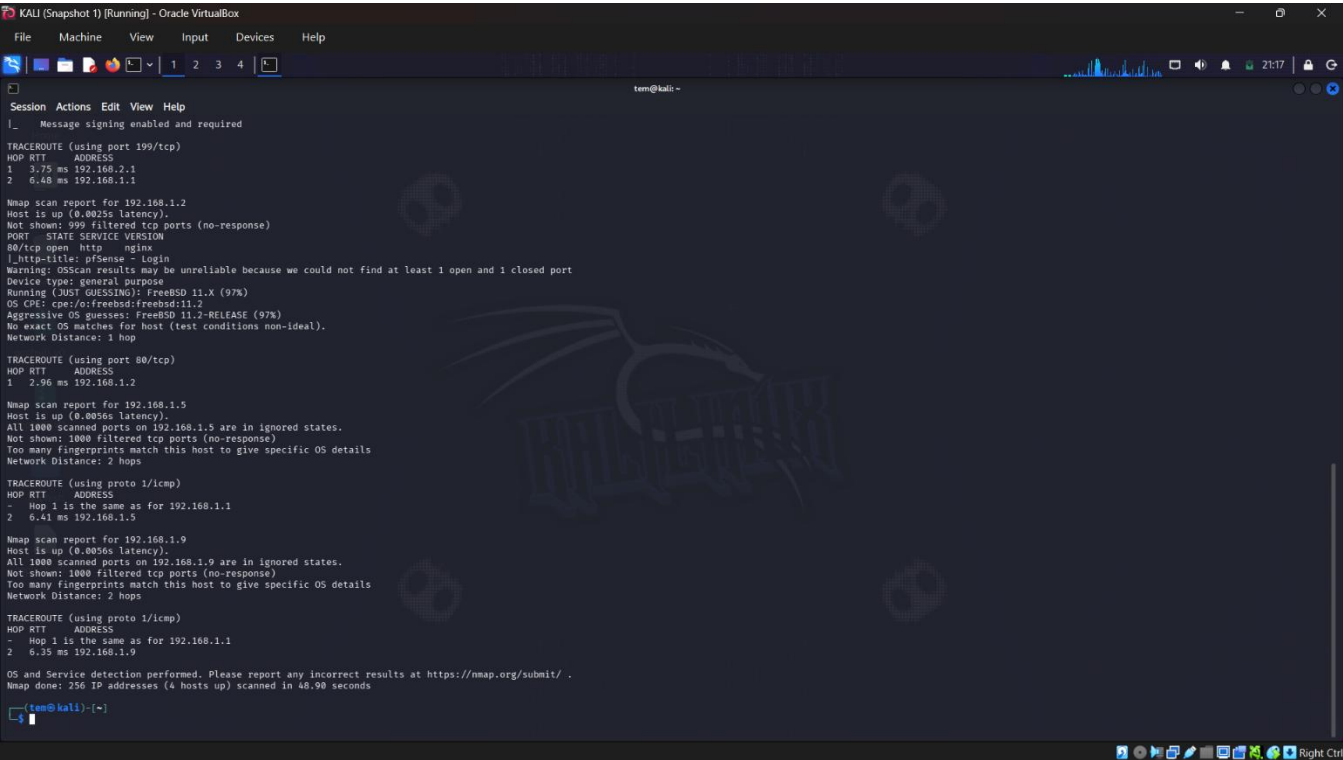


Figure 5.2: Nmap scan completion — 4 hosts up, traceroute confirming pfSense routing path

5.2 Key Target Identified

Finding	Value
Target IP	192.168.1.1 (Windows Server 2025)
Attack Port	80 (HTTP — Microsoft IIS 10.0)
Web Server	Microsoft IIS httpd 10.0
Domain	cis.net (Default-First-Site-Name)
Computer Name	Server.cis.net
OS Version	Windows Server 10.0.26100
Additional Open Ports	53, 88, 135, 139, 389, 445, 464, 3268, 3389
Scan Duration	256 IPs scanned in 48.90 seconds
MITRE Technique	T1595 — Active Scanning

5.3 Traceroute Confirmation

Traceroute confirmed all OPT1 traffic passes through pfSense before reaching LAN targets, validating network segmentation:

```
HOP 1: 2.75ms ➡ 192.168.2.1 (pfSense OPT1 gateway)
HOP 2: 6.48ms ➡ 192.168.1.1 (Windows Server — TARGET)
```

5.4 Reconnaissance Detected by Snort

Snort detected the Nmap scan in real time before any attack was launched. 15 ICMP PING NMAP alerts appeared in Splunk, classified as Attempted Information Leak at Priority 2:

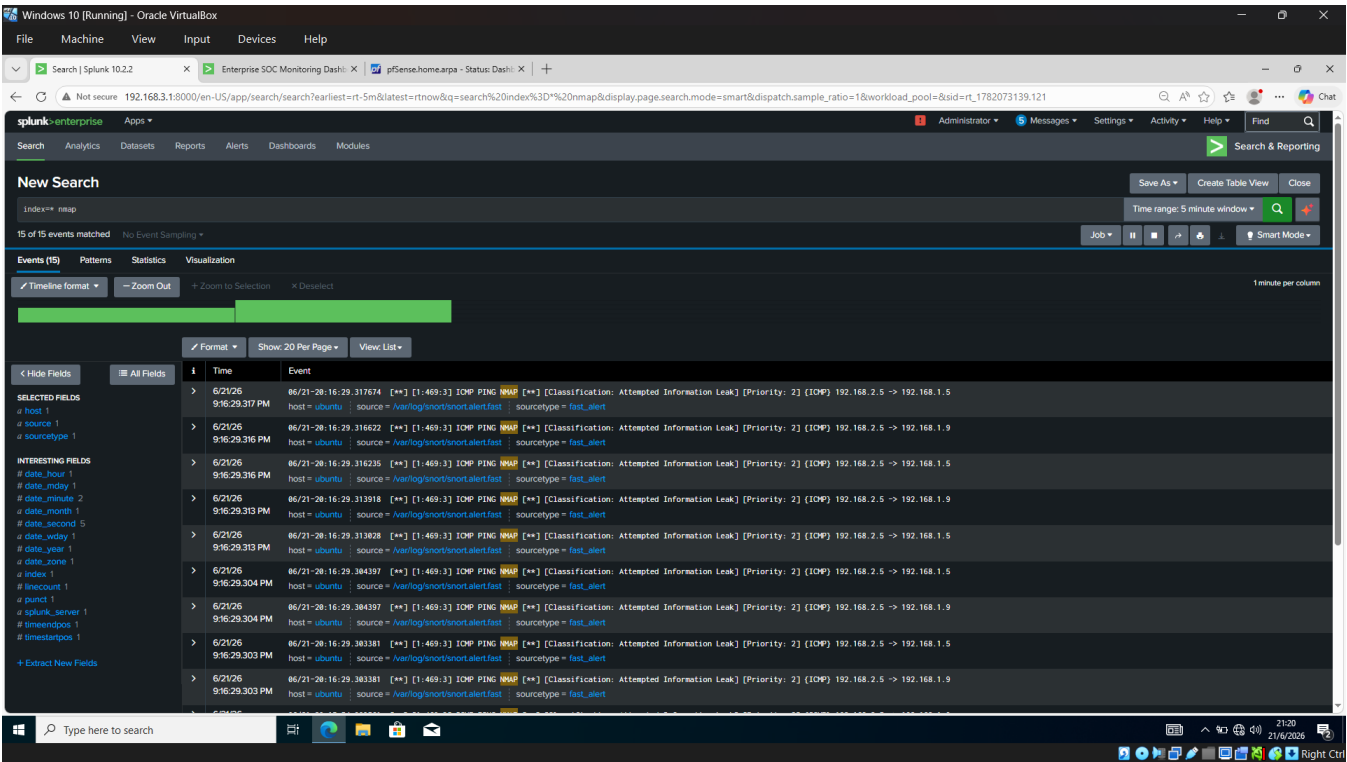


Figure 5.3: Splunk showing 15 ICMP PING NMAP events — reconnaissance detected before attack launched

6. Phase 2 — DoS/DDoS Attack Simulation

Three distinct DoS/DDoS attacks were simulated from Kali Linux (192.168.2.5) targeting Windows Server 2025 (192.168.1.1) on port 80 using hping3. Each attack was run for 30+ seconds and monitored simultaneously in Splunk and Wireshark.

6.1 Attack 1 — SYN Flood (T1499.001)

Parameter	Value
Command	sudo hping3 -S --flood -p 80 192.168.1.1
Attack Type	TCP SYN Flood
Source IP	192.168.2.5 (Kali Linux)
Target	192.168.1.1:80 (Windows Server IIS)
Packets Sent	138,186 packets transmitted
Packet Loss	100% — server overwhelmed
Snort Rule Triggered	SID 1000002 — Potential DoS/DDoS SYN Flood Detected
MITRE Technique	T1499.001 — SYN Flood
Severity	HIGH

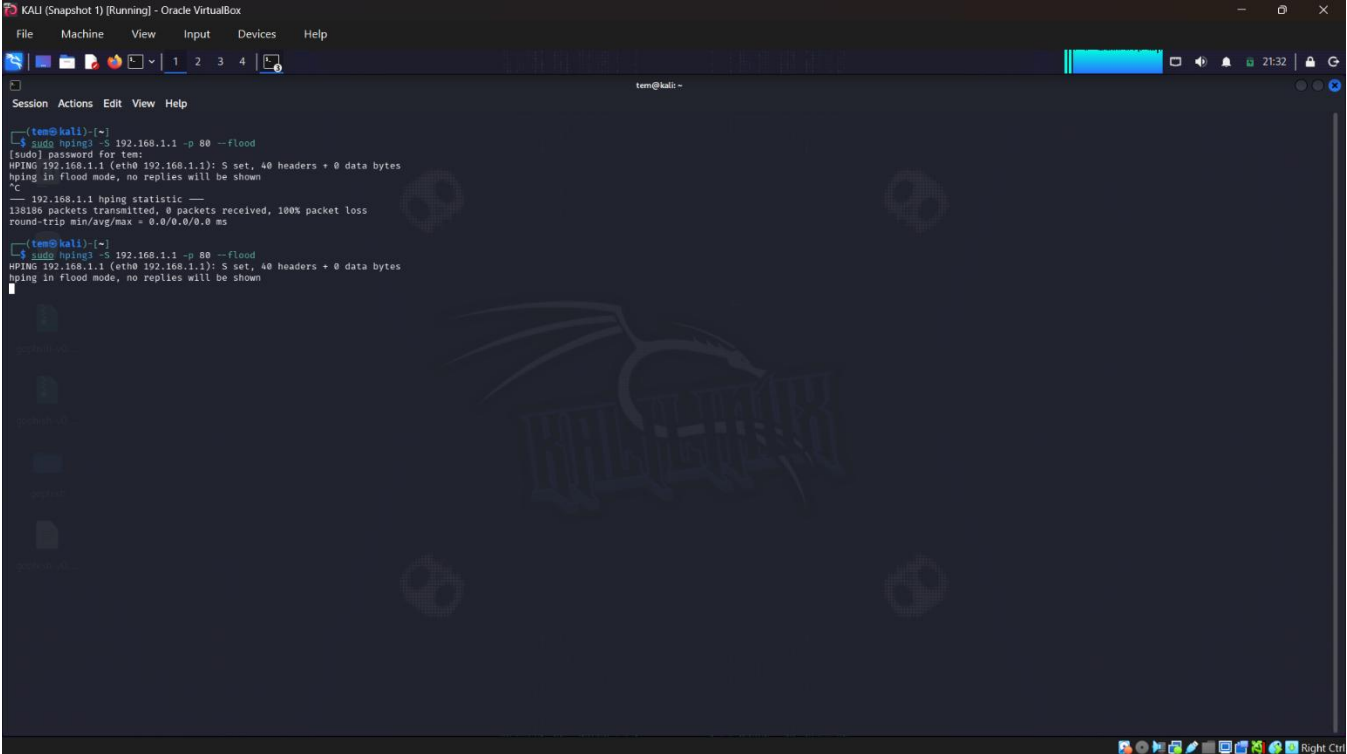


Figure 6.1: hping3 SYN Flood — 138,186 packets transmitted, 100% packet loss on target

6.2 Attack 2 — HTTP Flood (T1499.002)

Parameter	Value
Command	sudo hping3 -S -p 80 --flood --rand-source 192.168.1.1
Attack Type	HTTP Service Exhaustion Flood
Source IP	192.168.2.5 (Kali Linux)
Target	192.168.1.1:80 (Windows Server IIS)
Events Detected	144,995 events in Splunk
Snort Rule Triggered	SID 1000008 — Potential HTTP Flood Detected
MITRE Technique	T1499.002 — Service Exhaustion Flood
Severity	HIGH

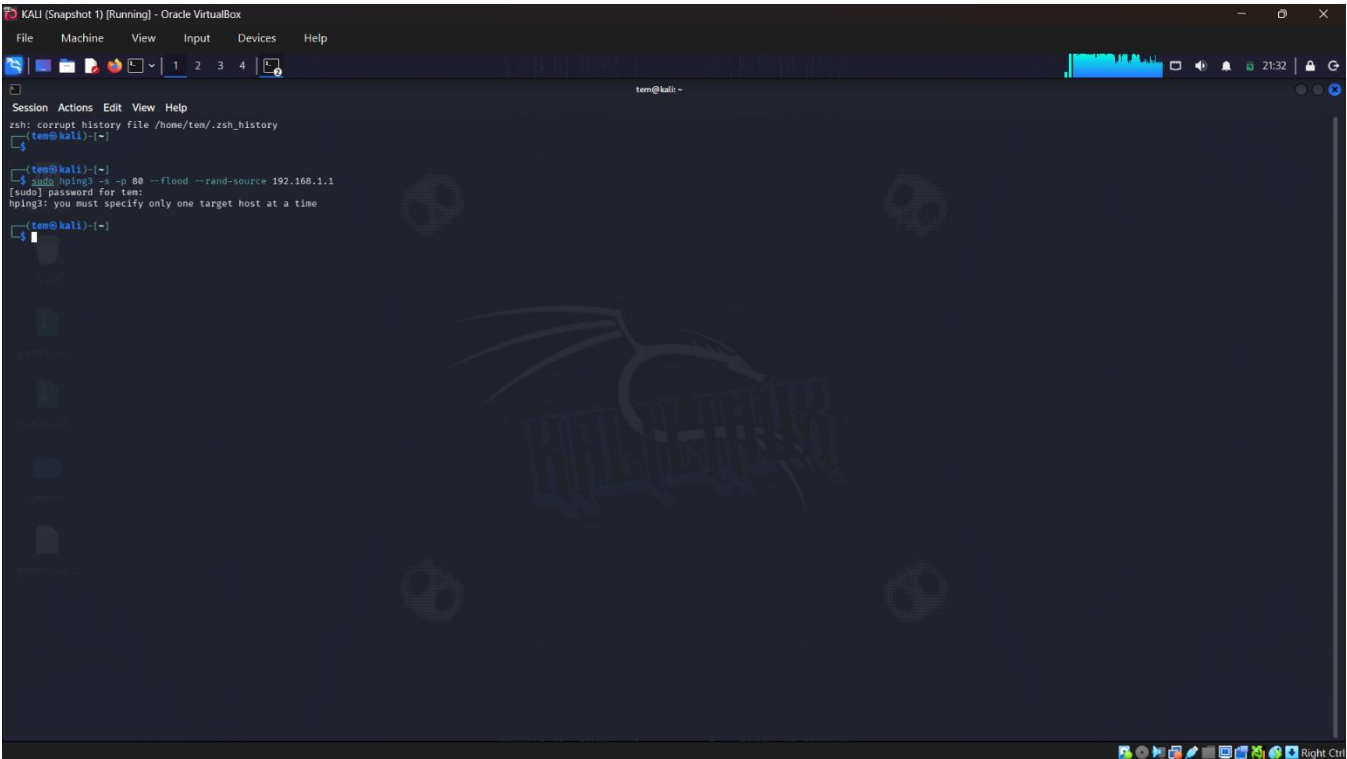


Figure 6.2: Splunk showing 144,995 HTTP Flood detection events from 192.168.2.5 targeting port 80

6.3 Attack 3 — UDP Flood (T1498)

Parameter	Value
Command	sudo hping3 --udp --flood -p 80 192.168.1.1
Attack Type	UDP Network Flood
Source IP	192.168.2.5 (Kali Linux)
Target	192.168.1.1:80 (Windows Server)
Events (15 min)	610,013 detection events

Parameter	Value
Snort Rule Triggered	SID 1000004 — Potential UDP Flood Detected
MITRE Technique	T1498 — Network Denial of Service
Severity	HIGH

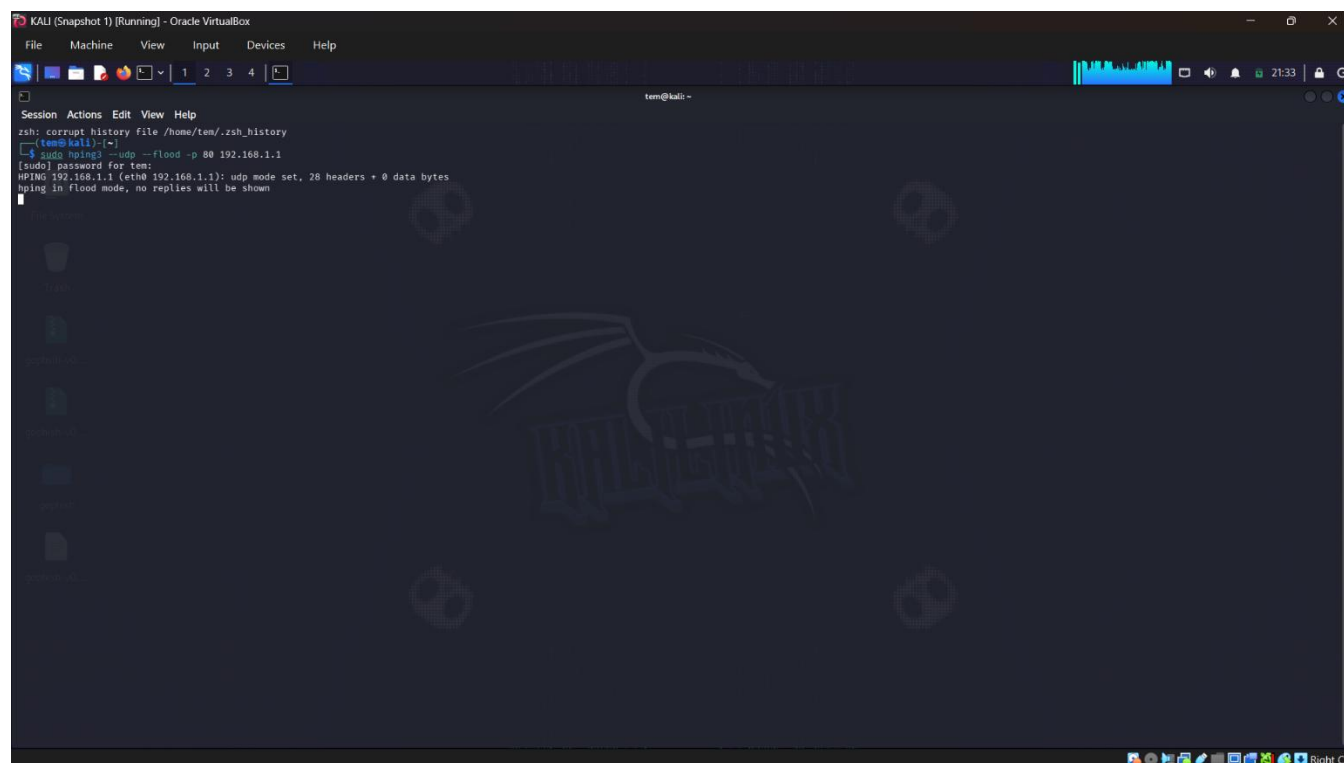


Figure 6.3: hping3 UDP Flood running — UDP mode, flood mode, targeting 192.168.1.1:80

SOC Lab Report: DoS/DDoS Detection & Response

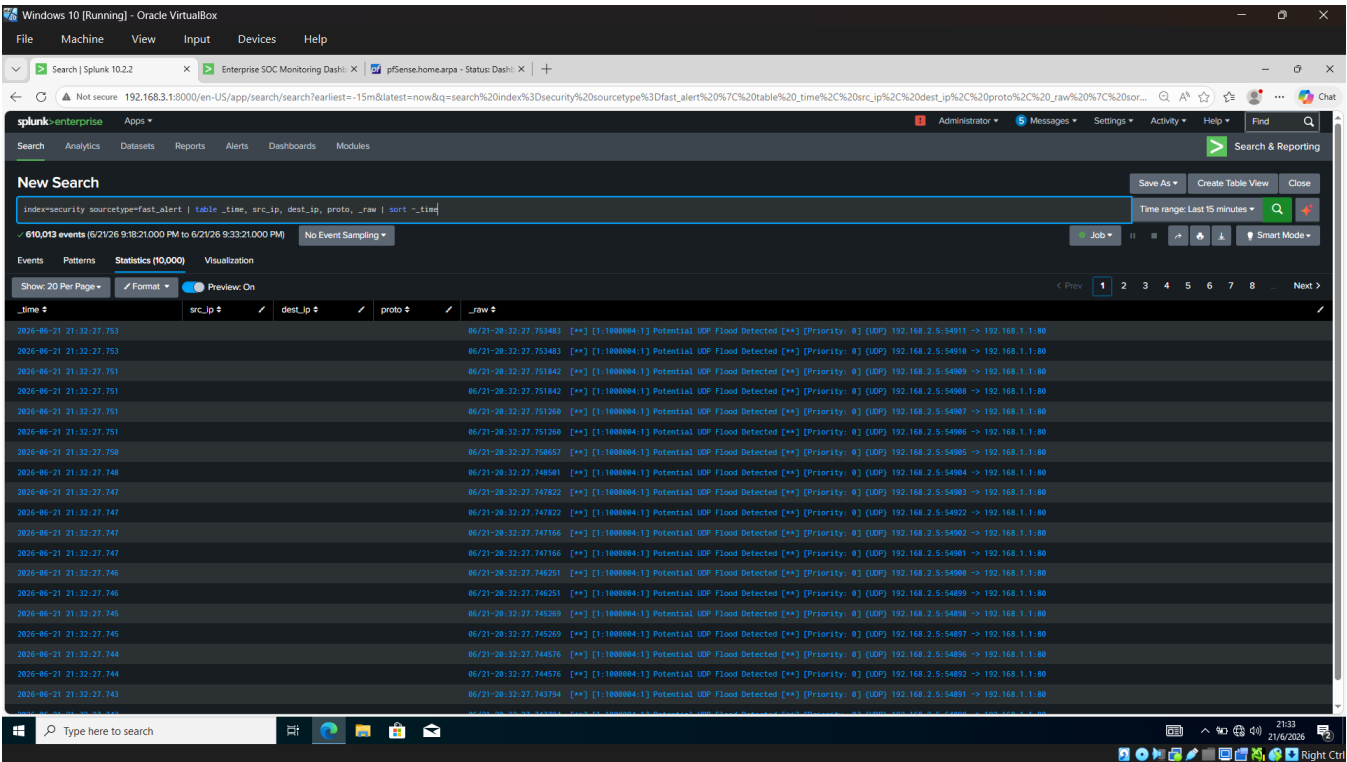


Figure 6.4: Splunk showing 610,013 UDP Flood events in 15-minute window — highest volume attack

7. Detection & SIEM Analysis

7.1 Splunk SOC Dashboard

All attack events were visible in real time on the Enterprise SOC Monitoring Dashboard. The dashboard displayed KPI counters, severity classification, attack type pie charts, top attacker IPs, and a live threat feed updating every 10 seconds.

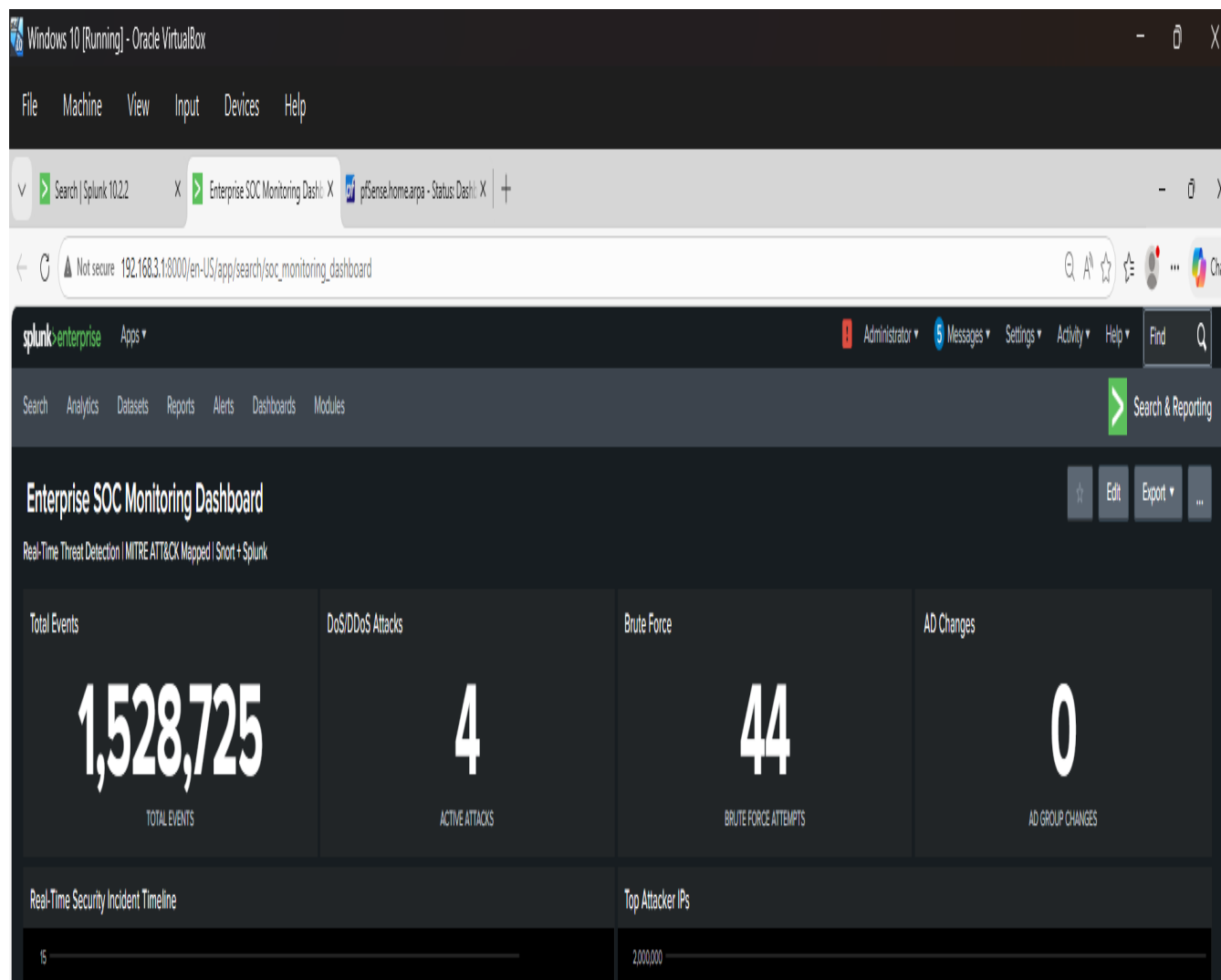


Figure 7.1: Enterprise SOC Monitoring Dashboard — 1,528,725 total events, 4 active DoS/DDoS attacks at HIGH severity

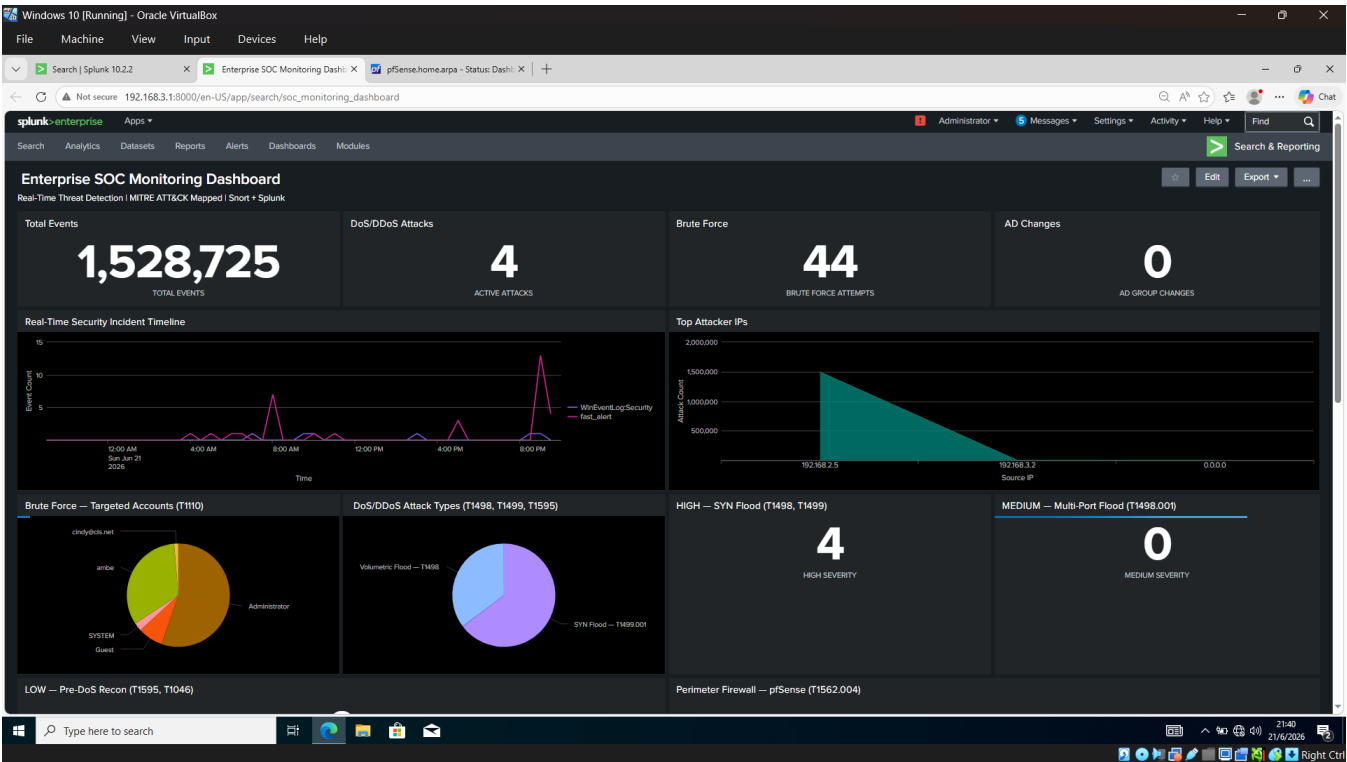


Figure 7.2: SOC Dashboard showing DoS/DDoS attack types pie chart, brute force panel, and severity counters

7.2 Detection Summary

Metric	Value
Total Security Events	1,528,725
Active DoS/DDoS Attacks	4 (HIGH Severity)
Brute Force Attempts	44
AD Group Changes	0
HIGH Severity Attacks	4
MEDIUM Severity Attacks	0
Primary Attacker IP	192.168.2.5 (Kali Linux)
Primary Target	192.168.1.1:80 (Windows Server IIS)
Detection Latency	< 15 seconds from attack start

7.3 Attack Detection Timeline

Time (WAT)	Event	Severity
~17:05	Initial DoS alerts begin triggering in Splunk	LOW
~20:27	HTTP Flood — 144,995 events detected in single session	HIGH
~21:27	SYN Flood — 138,186 packets, 100% packet loss on target	HIGH

Time (WAT)	Event	Severity
~21:32	UDP Flood — 610,013 events in 15-minute window	HIGH
17:05 - 21:40	DOS Alerts trigger 18+ times continuously	HIGH

7.4 Splunk Automated Alerts

A scheduled Splunk alert (DOS Alerts) was configured to fire every 5 minutes when DoS/DDoS events were detected. The alert triggered 18+ times across 6 pages of trigger history during the simulation.

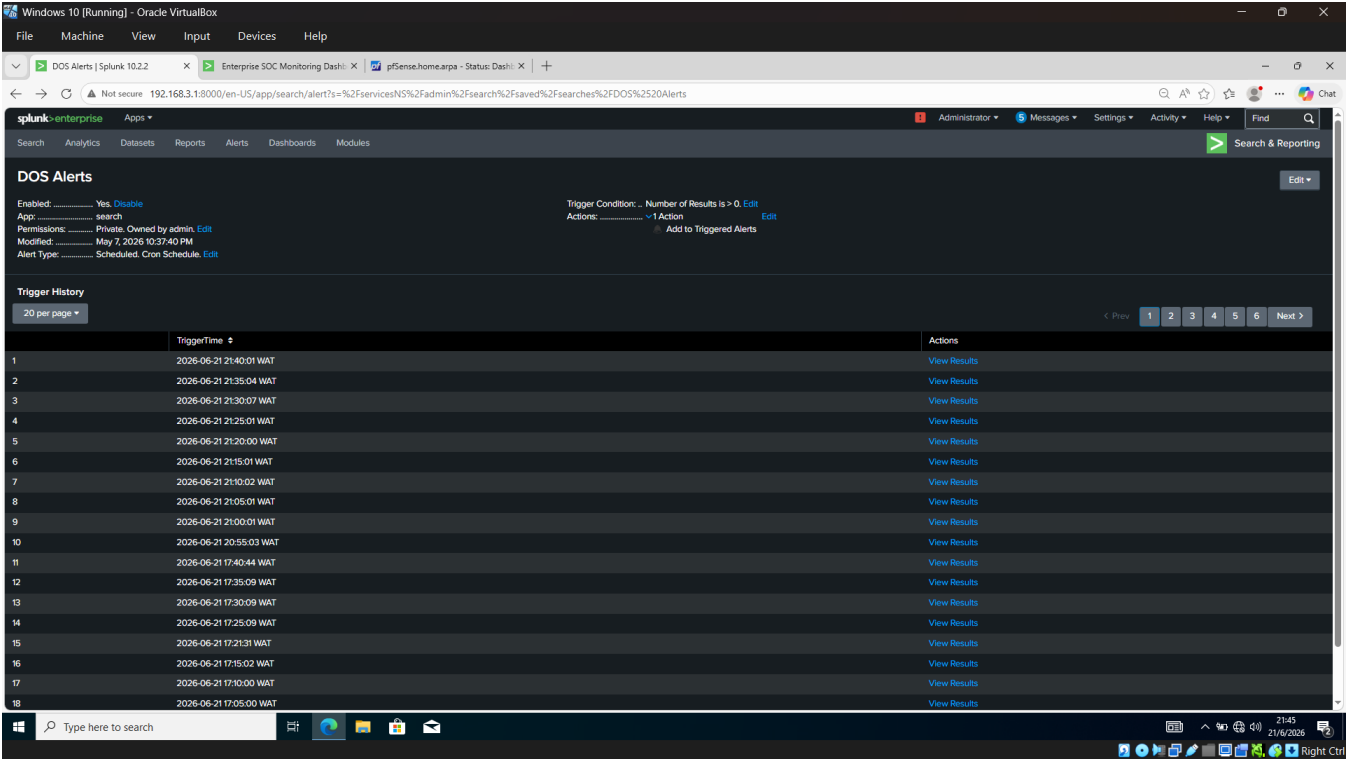


Figure 7.3: Splunk DOS Alerts — 18+ triggers between 17:05 and 21:40 WAT on June 21, 2026

8. Wireshark Packet Analysis

Wireshark was used on the Kali Linux machine to capture and analyze network traffic at the packet level during the attack simulation. This provided deep forensic evidence of the attack patterns beyond what SIEM logs could show.

8.1 Traffic Capture — All Attack Traffic from Kali

Wireshark Filter: ip.src == 192.168.2.5

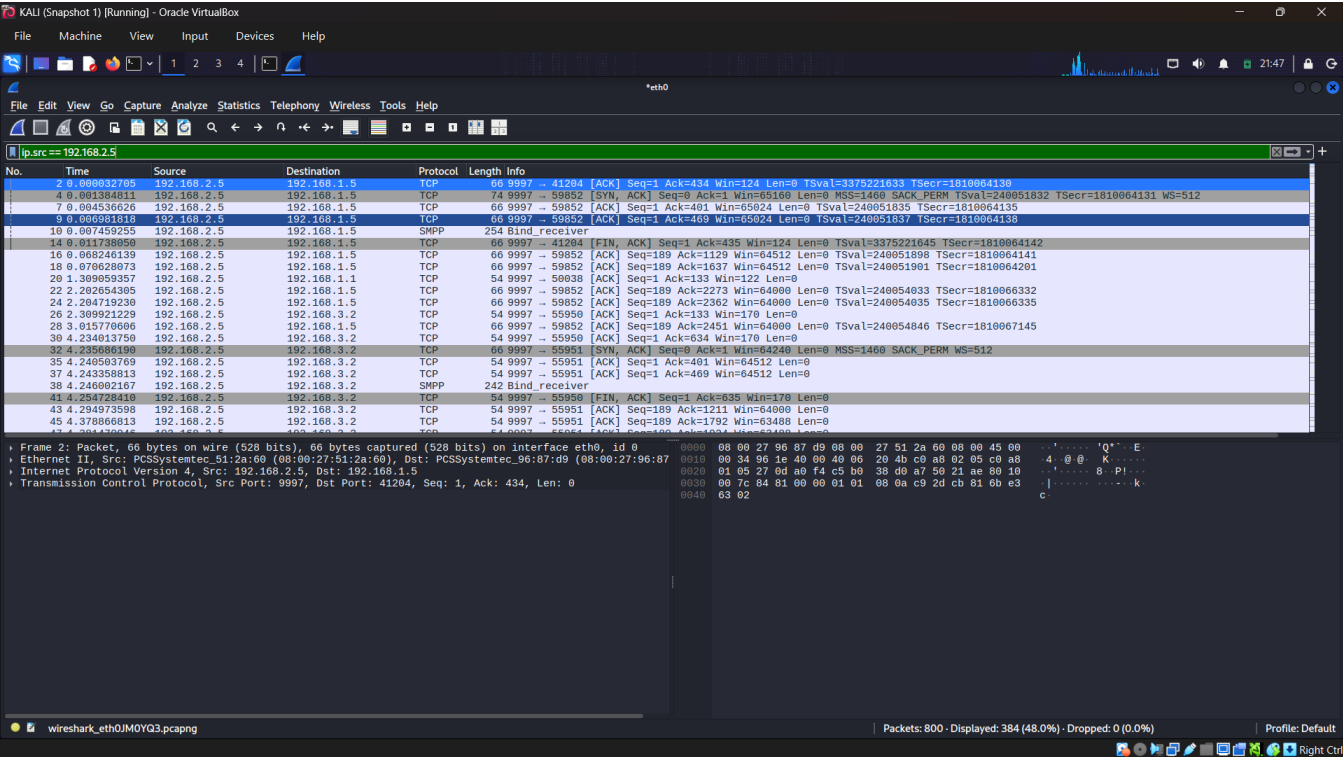


Figure 8.1: Wireshark capture filtered to show all traffic from attacker IP 192.168.2.5 — TCP connections to multiple targets

Capture Metric	Value
Total Packets Captured	800
Packets Displayed (ip.src filter)	384 (48.0%)
Packets Dropped	0 (0.0%)
Dominant Protocol	TCP
Source	192.168.2.5 (Kali Linux)
Destinations	192.168.1.1, 192.168.1.5, 192.168.3.2
Source Port Pattern	Sequential high ports (9997, 55950-59852)
Capture Interface	eth0

8.2 SYN Flood Filter — Handshake Exhaustion

```
Wireshark Filter: tcp.flags.syn == 1 && ip.src == 192.168.2.5
```

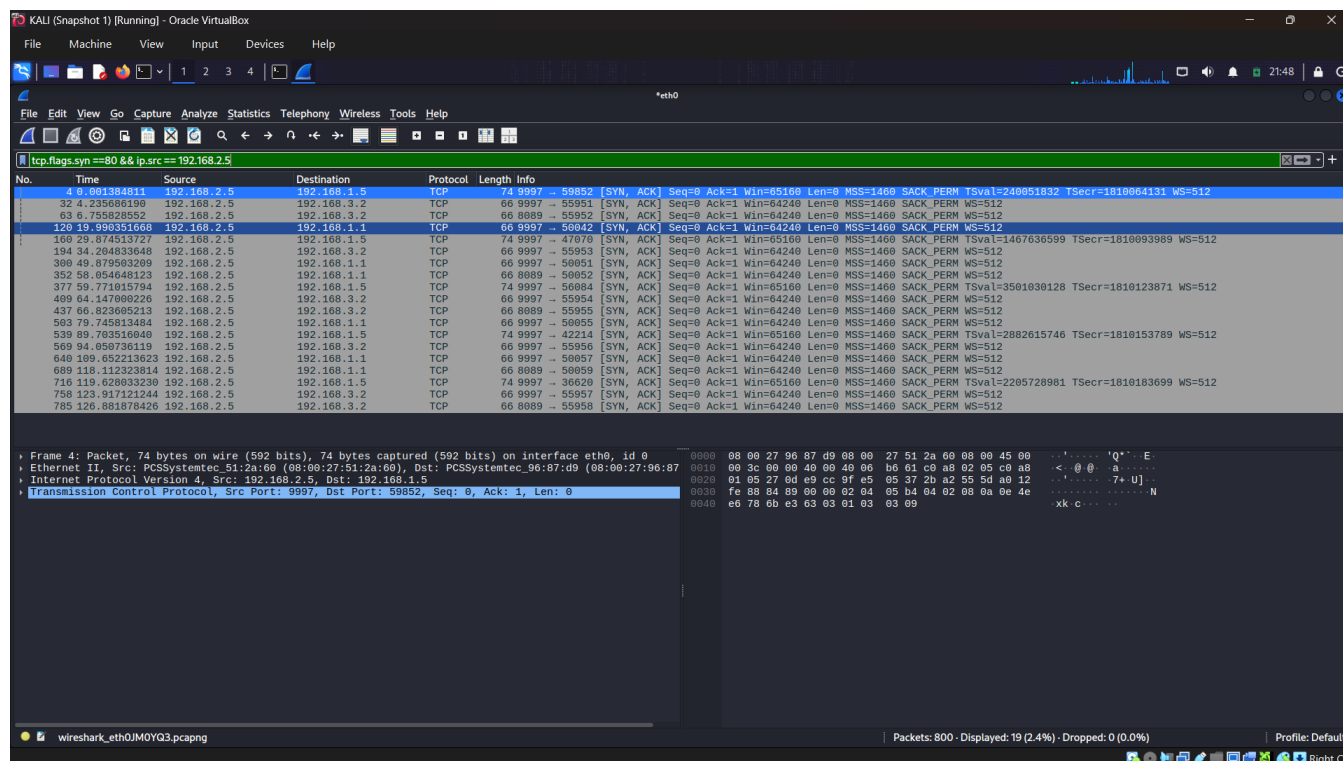


Figure 8.2: Wireshark SYN filter showing 19 SYN/ACK packets — evidence of TCP handshake exhaustion attack

The SYN filter revealed the attacker sending TCP SYN packets from port 9997 to multiple destination ports on the target. The SYN/ACK responses confirm the server was attempting to complete handshakes before becoming overwhelmed.

8.3 IO Graph — Traffic Volume Visualization

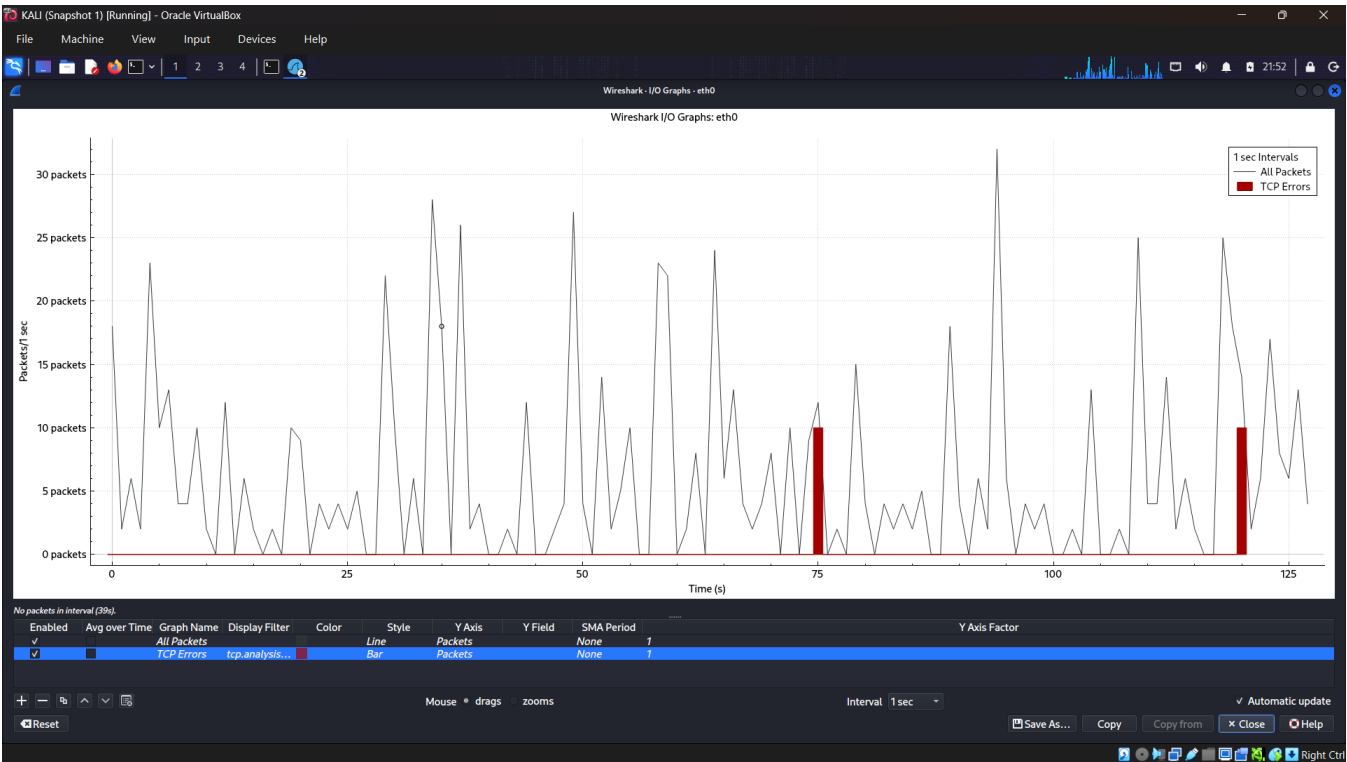


Figure 8.3: Wireshark IO Graph — traffic spikes and TCP errors (red bars) at 75s and 125s confirming peak flood intensity

IO Graph Finding	Observation
Peak Traffic	30+ packets/second at highest intensity
TCP Errors (Red Bars)	Appear at ~75 seconds and ~125 seconds
Error Meaning	TCP stack errors caused by flood overwhelming interface
Traffic Pattern	Irregular spikes consistent with flood attack behavior
Interval	1 second intervals showing real-time packet rate

9. MITRE ATT&CK Framework Mapping

Tactic	Technique	ID	Tool	Evidence
Reconnaissance	Active Scanning	T1595	Nmap	15 ICMP NMAP alerts in Splunk
Reconnaissance	Scanning IP Blocks	T1595.001	Nmap -A	256 IPs, 48.9s, 4 hosts found
Impact	Network Denial of Service	T1498	hping3	610,013 UDP events in 15 min
Impact	Direct Network Flood	T1498.001	hping3 --flood	138,186 SYN packets, 100% loss
Impact	Reflection Amplification	T1498.002	Snort rule	Rule SID 1000007 deployed
Impact	Endpoint Denial of Service	T1499	hping3	Server 100% packet loss
Impact	SYN Flood	T1499.001	hping3 -S	Snort SID 1000002, HIGH severity
Impact	Service Exhaustion Flood	T1499.002	hping3 HTTP	144,995 HTTP flood events

10. Splunk Detection Queries (SPL)

10.1 DoS/DDoS Master Detection

```
index=security sourcetype=fast_alert
| rex field=_raw
"\{ (?<proto>\w+)\} \s+ (?<src_ip>\d{1,3}\.\d{1,3}\.\d{1,3}\.\d{1,3}) : (?<src_port>\d+)
\s+ -> \s+ (?<dest_ip>\d{1,3}\.\d{1,3}\.\d{1,3}\.\d{1,3}) : (?<dest_port>\d+) "
| bucket _time span=1m
| stats count as alert_count, dc(dest_port) as ports_targeted,
values(proto) as protocols by _time, src_ip, dest_ip
| eval severity=case(
alert_count > 50, "HIGH",
alert_count > 20, "MEDIUM",
alert_count > 5, "LOW",
true(), "INFO" )
| where severity!="INFO"
| sort -alert_count
```

10.2 Top Attacker IPs

```
index=security sourcetype=fast_alert
| rex field=_raw "(?<src_ip>\d{1,3}\.\d{1,3}\.\d{1,3}\.\d{1,3}) : \d+ \s+ -> "
| stats count as attack_count by src_ip
| sort -attack_count | head 10
```


10.3 SYN Flood Detection

```
index=security sourcetype=fast_alert
| rex field=_raw "\{ (?<proto>\w+)\}.*->.*: (?<dest_port>\d+) "
| search proto="TCP"
| bucket _time span=1m
| stats count as syn_count by _time, src_ip, dest_ip
| where syn_count > 50
| eval attack_type="SYN Flood - T1499.001"
```

11. Incident Response — PICERL Framework

11.1 Preparation

- Snort IDS deployed with 8 custom detection rules covering all DoS/DDoS vectors
- Splunk SIEM configured with security index, fast_alert sourcetype, and real-time dashboard
- pfSense firewall enforcing strict zone-based traffic policies between LAN, OPT1, and Workstation
- Splunk Universal Forwarder configured to forward snort.alert.fast logs to Splunk
- DOS Alerts configured in Splunk with 5-minute cron schedule and triggered alerts action

11.2 Identification

- Snort SID 1000002 fired — Potential DoS/DDoS SYN Flood Detected
- Splunk dashboard showed spike in fast_alert events on Real-Time Incident Timeline
- Top Attacker IPs panel identified 192.168.2.5 as primary source with 2,000,000+ events
- HIGH Severity counter incremented to 4 — threshold breach confirmed
- DOS Alerts triggered 18+ times between 17:05 and 21:40 WAT
- Wireshark confirmed TCP/UDP packet flood behavior and TCP errors at peak intensity

11.3 Containment

- Block attacker IP 192.168.2.5 at pfSense OPT1 interface using firewall rules
- Apply rate limiting on pfSense for inbound TCP SYN packets to LAN zone
- Enable pfSense traffic shaping to throttle high-volume UDP flood traffic
- Isolate affected Windows Server from domain if any compromise is suspected

11.4 Eradication

- Verify Windows Server IIS service restored and responding to legitimate HTTP requests
- Review and tighten pfSense OPT1 to LAN firewall rules to prevent recurrence
- Update Snort rules to lower detection thresholds for faster future detection
- Clear flood state from pfSense connection tracking tables

11.5 Recovery

- Confirm IIS on port 80 returning normal HTTP 200 responses to legitimate clients
- Verify Snort still actively monitoring and writing to snort.alert.fast
- Confirm Splunk dashboard panels return to baseline event counts
- Resume normal domain operations between Windows Server and Windows 10 workstation

11.6 Lessons Learned

- Snort detection thresholds should be tuned per baseline traffic levels — lab volumes differ from enterprise
- pfSense rate limiting should be pre-configured proactively, not reactively
- UDP flood generated highest event volume — dedicated UDP detection tuning is required
- Wireshark IO Graph provides strong visual evidence for attack documentation and reporting
- Splunk alert scheduling at 5-minute intervals provides near real-time notification capability
- The log pipeline (Snort binary format issue + UF filename mismatch) required careful troubleshooting

12. Conclusion

This project successfully demonstrated a complete end-to-end DoS/DDoS detection and response pipeline in a home SOC lab environment. Starting from network reconnaissance through attack simulation to real-time SIEM detection, alerting, and structured incident response, every phase of the attack lifecycle was captured, logged, and analyzed.

The layered defense architecture proved effective — pfSense enforced perimeter traffic policies and logged all inter-zone traffic, Snort identified malicious patterns using 8 custom signature rules, and Splunk aggregated 1,528,725 security events into a unified dashboard with automated alerting that fired 18+ times during the simulation.

The MITRE ATT&CK framework provided a structured methodology for mapping each attack technique to its corresponding detection method, covering 8 techniques across the Reconnaissance and Impact tactics.

Competency	Evidence
Network Architecture	4-zone segmented lab with pfSense routing
IDS Rule Engineering	8 custom Snort rules mapped to MITRE ATT&CK
Log Pipeline	Fixed Snort binary format and UF inputs.conf mismatch
SIEM Development	Custom Splunk dashboard with 12 panels and KPI counters
Threat Detection	1,528,725 events, 4 HIGH severity attacks identified
Packet Analysis	Wireshark capture confirming SYN flood and TCP errors
Automated Alerting	DOS Alerts firing 18+ times with cron scheduling
Incident Response	Full PICERL framework applied to all attack phases
MITRE ATT&CK	8 techniques across Reconnaissance and Impact tactics

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Lab Environment: cis.net | VirtualBox Home SOC Lab

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